Biospectra : ISSN: 0973-7057, Vol. 19(2), September, 2024, pp. 179-186

An International Biannual Refereed Journal of Life Sciences

Plant Science



ISSN : 0973-7057

In vitro screening of stem extracts of *Cissus quadrangularis* L. for antibacterial activity against selected plant pathogen using different solvents.

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Received : 18th June, 2024 ; Revised : 16th July, 2024

DOI:-https://doi.org/10.5281/zenodo.15106364

Abstract- Medicinal plants are rich in pharmacologically active compounds, and *Cissus quadrangularis* L., a species from the Vitaceae family, is widely recognized for its therapeutic properties. Known for its succulent stems, this plant is capable of producing various bioactive molecules. The aim of this study was to evaluate the antibacterial activity of *Cissus quadrangularis* L. stem extract, prepared using different solvents, against selected plant pathogens. The antibacterial activity was assessed using the disc diffusion method, while the Minimum Inhibitory Concentration (MIC) was determined through the broth dilution method. The results showed that the dichloromethane extract exhibited the strongest antibacterial effect against *Xanthomonas campestris* (ZOI: 11.84±0.15) and *Ralstonia solanacearum* (ZOI: 12.28±0.12), whereas the hexane and acetone extracts demonstrated the weakest activity against *Xanthomonas campestris* (ZOI: 1.75±0.17) and *Ralstonia solanacearum* (ZOI: 6.45±0.15), respectively. In terms of MIC, the hexane extract showed the highest values against *Xanthomonas campestris* (MIC - 75 µg & IC_{50} - 46.77 µg), while the dichloromethane extract had the lowest (MIC - 40 µg & IC_{50} - 31.59 µg). For *Ralstonia solanacearum*, the acetone extract displayed the highest MIC values (MIC - 60 µg & IC_{50} – 38.63 µg), while the dichloromethane extract showed the lowest (MIC - 40 µg & IC_{50} - 31.04 µg). The positive results from this study support the traditional use of *Cissus quadrangularis* as an antibacterial agent against the tested plant pathogens.

Key words: Antibiotic, *Cissus quadrangularis, Xanthomonas compestris, Ralstonia solenoserum*

INTRODUCTION

Plants are a primary source of medicine for a large portion of the global population. One of the key advantages of medicinal plants is their safety, making them a preferred option for various treatments. They are often more affordable, effective, and readily available worldwide.[1] Traditional medicinal systems like Ayurveda, Unani, Siddha, and modern medicine all incorporate plant-based remedies to treat various ailments.[2] In the Indian subcontinent alone, a diverse range of medicinal plants are utilized for the treatment of numerous diseases.[3] While approximately 20,000 medicinal plants are recognized in Indian medicine, around 7,000 to 7,500 of these are actively used by traditional communities.[4,5] Phytochemical analyses of plants have identified a variety of bioactive compounds, including alkaloids, tannins, flavonoids, steroids, and glycosides, many of which serve as intermediates in the development of modern pharmaceuticals.[6]

Cissus quadrangularis is considered one of the most valuable plants due to its numerous medicinal benefits. It is a perennial herb from the grape family (Vitaceae), characterized by its stout, fleshy, quadrangular stems. This

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plant, also known as *Cissus succulent*, is commonly referred to as *horjora* in Hindi and *pirandai* in Tamil. It is native to regions such as India, Sri Lanka, Malaysia, Java, and parts of West Africa. *Cissus quadrangularis* thrives in tropical forest areas across Asia and Africa and is widely recognized in India for its medicinal properties.[7] This evergreen climber grows rapidly in these regions, making it an easily accessible resource for traditional medicine. *Cissus quadrangularis* has been extensively studied for its phytochemical composition, pharmacological properties, and toxicological effects. The plant extract contains various bioactive compounds, including alkaloids, tannins, lignin, suberin, phenols, flavonoids, resveratrol, piceatannol, pallidol, perthenocissin, phytosterols, and others. Among these, ascorbic acid, triterpenes, beta-sitosterol, ketosterols, two asymmetrical tetracyclic triterpenoids, and calcium have been identified as key phytochemicals.[8,9] The plant exhibits a range of therapeutic properties, such as antimicrobial, antioxidant, anti-inflammatory, anticancer, and cytotoxic effects. Additionally, *Cissus quadrangularis* is commonly used in traditional medicine for promoting bone healing.[10-13]

Antimicrobial compounds derived from plants may inhibit bacterial growth through mechanisms different from those of conventional antimicrobials, potentially offering significant clinical benefits in treating microbial infections.[14] The discovery of penicillin from *Penicillium notatum* by Alexander Fleming in 1929 marked the beginning of the antibiotic era, and since then, thousands of antibiotics have been identified from various plant sources. *Cissus quadrangularis* has shown notable antimicrobial activity in several studies. Research has demonstrated that this plant exhibits antimicrobial effects against a variety of plant pathogens, including *Bacillus subtilis*, *Bacillus cereus*, *Staphylococcus aureus*, *Streptococcus* species, *Proteus mirabilis*, *Pseudomonas aeruginosa*, *Geotrichum candidum*, *Mucor*, *Aspergillus niger*, *Fusarium proliferatum*, and others.[15-17]

MATERIALS & METHODS

Collection and identification of plant sample

Fresh stems of *Cissus quadrangularis* L. were collected from the Supaul district in Bihar, located at a latitude of 26.5520640 and longitude of 87.0555330. The plant and its stems were authenticated by Professor Rimjhim Sheel, Principal of GDM College, Patliputra University, Patna. A voucher specimen of the plant has been deposited at the University Department of Botany, Patliputra University, Patna, Bihar, India, for future reference. To prepare the plant material, the stems were first washed thoroughly with tap water, followed by a rinse with distilled water. The cleaned stems were then shade-dried and ground into a fine powder, which was stored in a sealed, clean container for later use.

Extraction of plant sample

The stems of *Cissus quadrangularis* L. were extracted using the Soxhlet extraction method.[18,19] A total of 50 g of dried powdered stems were placed in the thimble of a Soxhlet apparatus, and 350 ml of different solvents-methanol, acetone, chloroform, dichloromethane, hexane, and aqueous were used in sequence. These solvents helped dissolve the active compounds from the plant material, leaving the residual stems as a precipitate. The extraction process continued until the solvent in the thimble became clear, which usually took about 8 hours. The final extract was then dried in a water bath, resulting in a dark orange residue.

ANTIBACTERIAL ASSESSMENT

Plant pathogen (bacterial) samples

Bacterial strains were obtained from the Microbial Type Culture Collection (MTCC) at the Indian Agricultural Research Institute (IARI), New Delhi. The strains included *Xanthomonas campestris* (BH0001) and *Ralstonia solanacearum* (BI0001).

Screening of antimicrobial property by disc diffusion Test

For the antibacterial activity of the selected plant extracts, microorganisms such as *Xanthomonas campestris* and *Ralstonia solanacearum* were used. The antibacterial assay was conducted using the disc diffusion method, as described in previous studies.[20,21] Briefly, uniform discs of equal size were punched out from Whatman No. 1 filter paper using a punching machine. These discs were then impregnated with the crude plant extract and reference antibiotics.[22]

Preparation of Nutrient Agar:

14 gms of synthetic nutrient agar were dissolved in 500 ml of distilled water. Two spatulas of agar were added to the solution, which was then microwaved for 2 minutes. After boiling, the mixture was autoclaved at 121ºC for 15 minutes.

Preparation of Inoculum:

Five milliliters of saline water were placed in a test tube and sterilized by autoclaving for 15 minutes at 121ºC.

A loop of each bacterial isolate was then inoculated into the test tube and mixed thoroughly using a vortex shaker.

Inoculation of plates:

0.1 ml of each bacterial suspension was pipetted onto the plates and evenly spread over the surface of the medium using a sterilized cotton swab.

Loading of disc on plate:

The sterilized Petri plates were left at room temperature for 15 minutes to allow the surface moisture to evaporate before placing the plant extract and reference antibiotic-impregnated discs. Using sterile forceps, the discs were carefully placed on the plates under aseptic conditions. The plates were then incubated at 37ºC for 24 hours, within 15 minutes of disc placement.

Measurement of Zone of Inhibition

After the incubation period, the zone of inhibition (ZOI) was observed. The plates were inverted onto a black, non-reflective surface. A measuring scale was placed behind the inverted plate, and the margins of the zone of inhibition were measured to determine its size.

Minimum Inhibitory Concentration (MIC):

The antibacterial efficacy (MIC) of the stem extracts was determined using the broth dilution method.[23,24] The bacterial cultures were freshly inoculated in Nutrient Broth (NB) and incubated for 24 hours in a shaking incubator at 37°C and 150 rpm. A 20 µl aliquot of the overnight culture was added to 5 mL of Nutrient Broth containing different concentrations of plant extracts. A plant extract-free medium was used as a control. All samples were incubated for 24 hours at 37°C and 150 rpm. The absorbance was measured at 600 nm using a spectrophotometer, and the percentage antibacterial activity was calculated using the following formula:

$$\% \text{ inhibition} = \frac{OD(contrl) - \text{OD(sample)}}{OD(control)} \times 100$$

Where,

Control: - NB + bacterial inoculum without plant extracts

Sample: - NB + bacterial inoculum with plant extracts

RESULT & DISCUSSION

Antibacterial sensitivity test by Disc diffusion method

As mentioned earlier, several studies have highlighted the various pharmacological activities of *Cissus quadrangularis* L. including its antimicrobial properties.[12,13,25,26] Stem extracts of *Cissus quadrangularis L.* demonstrated antimicrobial efficacy against nearly all tested organisms. The mechanism behind the antibacterial activity of these extracts may involve their attachment to the bacterial cell membrane, disrupting permeability, osmoregulation, electron transport, and respiration.[27] Various methods are used to evaluate antimicrobial properties, including microtiter plate-based assays[28] and the disc diffusion (Kirby–Bauer) test[29]. In this study, the disc diffusion method was employed. To assess the antibacterial sensitivity of different stem extracts of *Cissus quadrangularis* L. a concentration-dependent response (15 µg/ml) was evaluated based on the zone of inhibition against *Ralstonia solanacearum* (Fig. 01, 02, and Table 01) and *Xanthomonas campestris* (Fig. 03, 04, and Table 01), with tetracycline and ampicillin used as reference antibiotics. Importantly, The Zone of inhibition is maximum for Dichloromethane extract (12.28±0.12 mm) and minimum for Acetone extract (6.45±0.15 mm) as compare to standard tetracycline (13.57±0.01) and ampicillin (14.85±0.01) against *Ralstonia solanacearum* (Fig. – 01, 02 and Table - 01). The order of Zone of inhibition of different extracts are Dichloromethane > Methanol > Hexane > Chloroform > Aqueous > Acetone. Similarly, The Zone of inhibition is maximum for Dichloromethane extract (11.84±0.15 mm) and minimum for Hexane extract (1.75±0.17 mm) as compare to standard tetracycline (13.57±0.01) and ampicillin (14.85±0.01) against *Xanthomonas campestris* (Fig. – 03, 04 and Table – 01).The order of Zone of inhibition of different extracts are Dichloromethane > Chloroform > Acetone > Aqueous > Methanol > Hexane. Overall, the dichloromethane extracts having maximum antibacterial activity against both the tested pathogens, i.e. *Xanthomonas compestris* and *Ralstonia solanacearum* while hexane and acetone extracts having minimum against *Xanthomonas compestris* and *Ralstonia solanacearum* respectively.

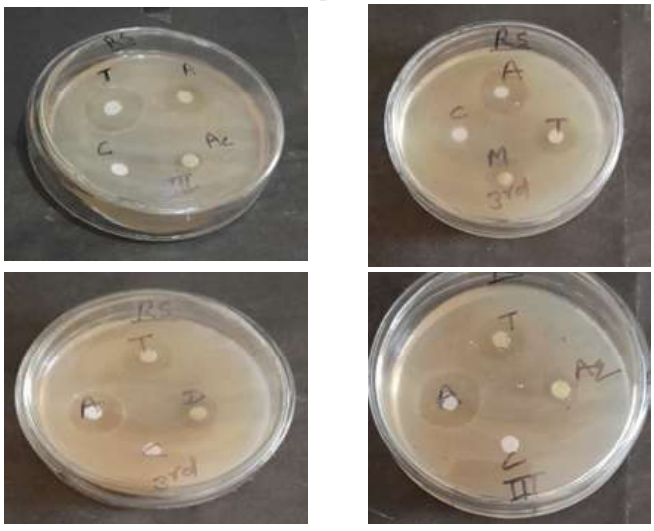




Fig. 1- Showing the zone of inhibition of different extracts against *Ralstonia solanacearum.*
Ac – Acetone, M – Methanol, D – Dichloromethane, Aq – Aqueous, Ch – Chloroform, H – Hexane, A – Ampicillin, T – Tetracycline, C – Control and RS - *Ralstonia solanacearum.*

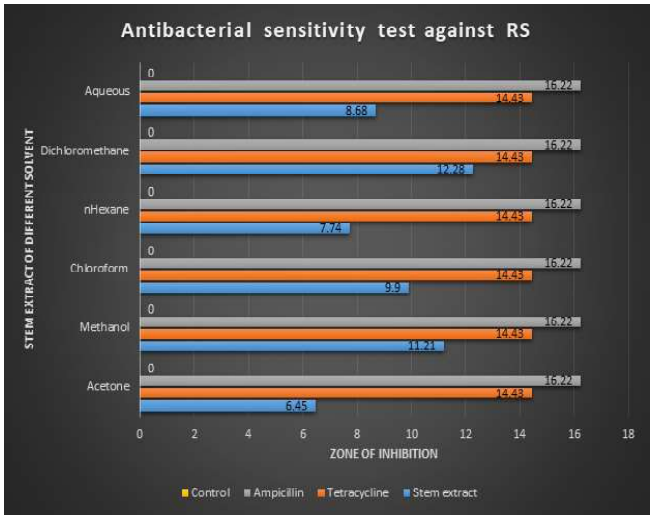


Fig. 2- Showing comparative analysis of Zone of inhibition of different stem extract of the plant *Cissus quadrangularis* against *Ralstonia solanacearum*.

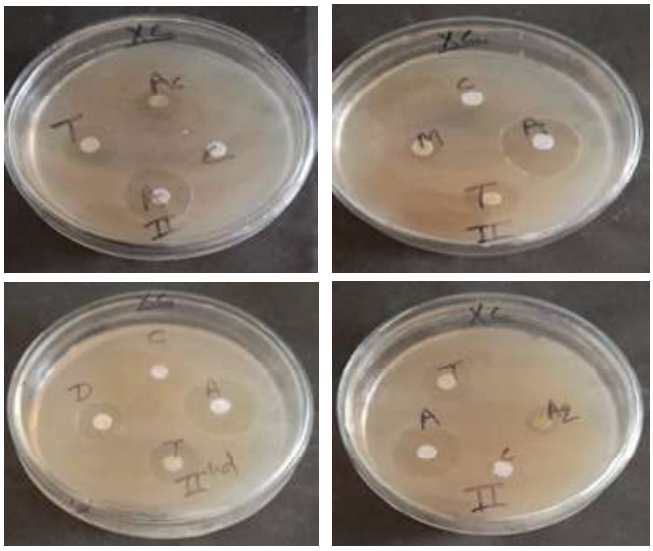




Fig. 3- Showing the zone of inhibition of different extracts against *Xanthomonas compestris.*
Ac – Acetone, M – Methanol, D – Dichloromethane, Aq – Aqueous, Ch –Chloroform, H – Hexane, A – Ampicillin, T – Tetracycline, C – Control and XC - *Xanthomonas compestris.*

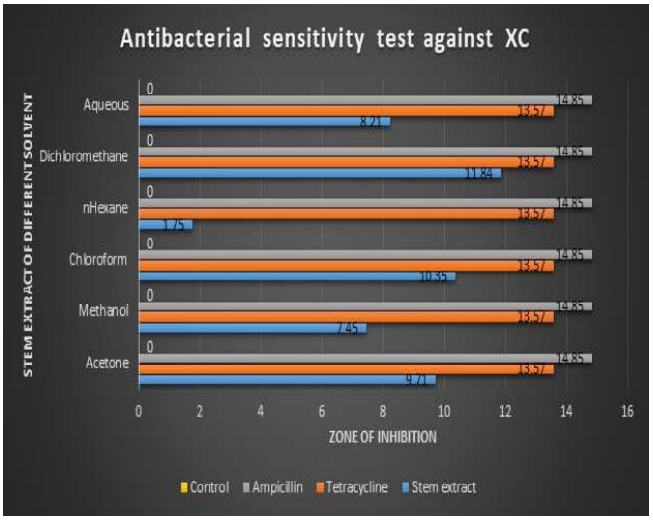


Fig. 4- Showing Comparative analysis of Zone of inhibition of different stem extract of the plant *Cissus quadrangularis.* against *Xanthomonas compestris*.

Table 1- Comparative analysis of Zone of inhibitions against *Xanthomonas compestris* and *Ralstonia solanacearum* developed by different solvent extract of the plant CQ.

Name of different solvent	Zone of inhibition against following bacteria	
	Xanthomonas compestris	*Ralstonia solanacearum*
Acetone	9.71±0.10	6.45±0.15
Methanol	7.45±0.17	11.21±0.12
Chloroform	10.35±0.15	9.90±0.19
Hexane	1.75±0.17	7.74±0.20
Dichloromethane	11.84±0.15	12.28±0.12
Aqueous	8.21±0.09	8.68±0.16
Tetracycline	13.57±0.01	14.43±0.01
Ampicillin	14.85±0.01	16.22±0.01
*Control	00.00	00.00

(*control- disc of respective solvents)

Determination of Minimum Inhibitory Concentration (MIC): -

The Minimum Inhibitory Concentration (MIC) values of the plant extract of different solvents (acetone, aqueous, chloroform, methanol, dichloromethane and hexane) of the plant *Cissus quadrangularis* against *Xanthomonas compestris* are maximum for hexane extract (MIC - 75 µg & IC_{50} - 46.77 µg) and minimum for dichloromethane extract (MIC - 40 µg & IC_{50} - 31.59 µg) (Fig. – 05 & Table - 02). The order of MIC values of different solvent extracts are in order Hexane > Methanol > Aqueous > Acetone > Chloroform > Dichloromethane. Similarly the Minimum Inhibitory Concentration (MIC) values of the plant extract of different solvents (acetone, aqueous, chloroform, methanol, dichloromethane and hexane) of the plant *Cissus quadrangularis* against *Ralstonia solanacearum* are maximum for acetone extract (MIC - 60 µg & IC_{50} – 38.63 µg) and minimum for dichloromethane extract (MIC - 40 µg & IC_{50} - 31.04 µg) (Fig. – 06 & Table - 02). The order of MIC values of different solvent extracts are Acetone > Aqueous > Chloroform > Hexane > Methanol > Dichloromethane. Overall, the dichloromethane extracts having minimum MIC value against both the tested pathogens, i.e. *Xanthomonas compestris* and *Ralstonia solanacearum* while hexane and acetone extracts having maximum against *Xanthomonas compestris* and *Ralstonia solanacearum* respectively it means dichloromethane extracts show maximum antibiotic activity at lower concentration as compare to other examined plant extracts against both pathogens.

Table 2- Comparative estimation of minimum inhibitory concentration (MIC) with IC_{50} value of plant extract of different solvents of *Cissus quadrangularis* against *Xanthomonas compestris* and *Ralstonia solanacearum*.

S.N.	Name of solvents	*Xanthomonas compestris*		*Ralstonia solanacearum*	
		MIC (in µg/ml)	IC_{50} (in µg/ml)	MIC (in µg/ml)	IC_{50} (in µg/ml)
01	Acetone	50	36.20	60	38.63
02	Methanol	60	39.24	40	31.34
03	Chloroform	45	34.73	50	36.29
04	Hexane	75	46.77	50	36.05
05	Dichloromethane	40	31.59	40	31.08
06	Aqueous	55	37.60	55	34.75

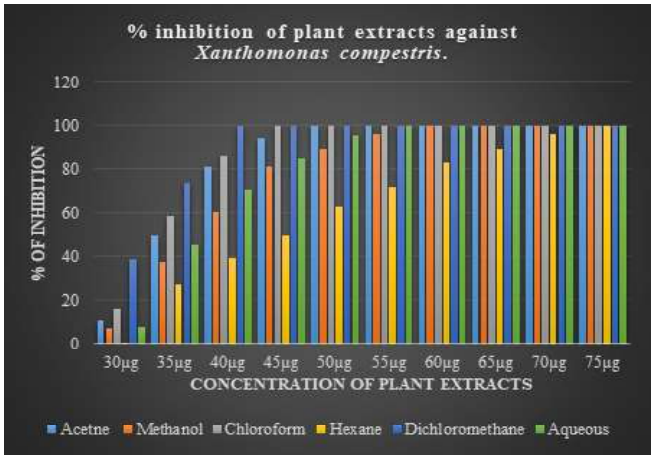


Fig. 5- Showing the percentage inhibition values at different concentration of the stem extracts of *Cissus quadrangularis* solvent against *Xanthomonas compestris.*

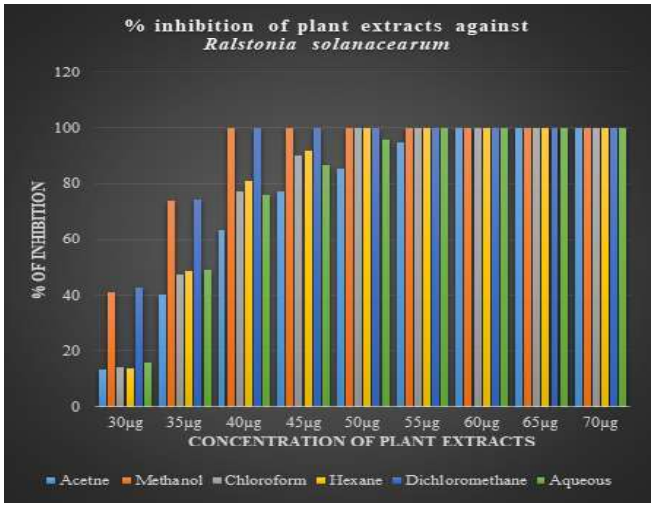


Fig. 6- Showing the percentage inhibition values at different concentration of the stem extracts of *Cissus quadrangularis* against *Ralstonia solanacearum*.

CONCLUSION

The plant *Cissus quadrangularis* L. evaluated in this study, is traditionally used to treat various ailments. The positive findings from this research provide scientific evidence supporting its traditional use as an antibacterial agent against the tested plant pathogens. This could offer farmers a biologically-based, environmentally friendly solution for managing crop diseases caused by these pathogens. Ultimately, the results of this study highlight the antibacterial properties of *Cissus quadrangularis* and provide valuable evidence supporting its use in folk medicine.

ACKNOWLEDGEMENT

We wish to thank HOD, Botany and Principal T.P.S. College, PPU and HOD, University department of Botany, PPU for their support and encouragement.

CONFLICT OF INTEREST

We hereby declare that there is no conflict of interest related to this study. The work presented in this paper is entirely original and was carried out with the highest standards of academic integrity. All research, data collection, and analysis were conducted by the authors, and the findings have not been influenced by any external parties or financial interests.

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